

The Real Business Cycle Model: Theory, Estimation, and Simulation

1 Historical Context

1.1 Real Business Cycle Theory and the Lucas Critique

Dynamic Stochastic General Equilibrium (DSGE) models emerged in the 1980s as a direct response to the **Lucas Critique (1976)**. Prior to the 1970s, macroeconomic analysis relied heavily on aggregate models like IS-LM and the Phillips Curve (AD-AS models). These models treated macroeconomic relationships as static.

Following the 1970s stagflation, the stable trade-off of the Phillips curve collapsed. Robert Lucas argued that traditional models could not evaluate policy changes because their parameters were not "structural"; they shifted as agents updated their expectations in response to new policies.

1.2 Neoclassical vs. New Keynesian

By the 1980s, the macroeconomic landscape was divided into two main schools regarding DSGE modeling:

1. Neoclassical / Real Business Cycle (RBC) Theory:

- **Flexible prices and wages:** Markets clear continuously (Walrasian equilibrium).
- **Real Shocks:** Fluctuations are driven by real supply-side shocks (primarily TFP/productivity), not market failures.
- **Neutrality of Money:** Nominal variables do not affect real outcomes in the short or long run.
- **Efficient Cycles:** Business cycles are efficient, optimal responses by agents to exogenous shocks.

The culmination of this theory was presented by **Kydland and Prescott (1982)**, who built the first fully articulated RBC model.

2. New Keynesian Theory:

- Retains the microfoundations and rational expectations of RBC models but introduces **nominal rigidities** (e.g., Calvo pricing, sticky wages).
- Because prices adjust slowly, monetary policy has real effects and can stabilize the economy.
- Demand matters: output gaps and consumption smoothing drive short-run dynamics via the New Keynesian Phillips Curve and the dynamic IS curve.

2 The Real Business Cycle Model Framework

The RBC model is a stochastic extension of the Ramsey-Cass-Koopmans (RCK) neoclassical growth model. By the First Welfare Theorem, in the absence of distortions, the decentralized competitive equilibrium is equivalent to the Social Planner's problem.

2.1 Preferences

Households maximize expected intertemporal utility over an infinite horizon:

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, L_t)$$

where $\beta \in (0, 1)$ is the discount factor. The utility function must satisfy the Inada conditions to ensure a preference for consumption smoothing (Permanent Income Hypothesis). We adopt a Constant Relative Risk Aversion (CRRA) utility function. Assuming logarithmic utility for both consumption and leisure to guarantee a balanced growth path, we have:

$$U(C_t, 1 - N_t) = \log(C_t) + \theta \log(1 - N_t)$$

where N_t represents hours worked (intensive margin), $L_t = 1 - N_t$ is leisure, and θ is the preference weight for leisure.

2.2 Technology

Production is governed by a Cobb-Douglas function subjected to a stochastic technology shock A_t :

$$Y_t = A_t K_{t-1}^\alpha N_t^{1-\alpha}$$

K_{t-1} denotes the predetermined capital stock chosen at $t - 1$ and used for production at t . The shock A_t evolves as an AR(1) process in logs:

$$\log(A_t) = (1 - \rho_a) \log(\bar{A}) + \rho_a \log(A_{t-1}) + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2)$$

2.3 Resource Constraint and Capital Accumulation

The economy-wide resource constraint includes an exogenous stochastic government expenditure G_t :

$$Y_t = C_t + I_t + G_t$$

Government spending follows a similar AR(1) process. Capital accumulates via the standard perpetual inventory method:

$$K_t = I_t + (1 - \delta)K_{t-1}$$

To prevent Ponzi schemes, the model imposes a **Transversality Condition (TVC)**:

$$\lim_{t \rightarrow \infty} \beta^t \lambda_t K_t = 0$$

3 Solving the Competitive Equilibrium

3.1 Firm's Problem

Firms operate in perfect competition, maximizing profits period-by-period:

$$\max_{K_{t-1}, N_t} \Pi_t = A_t K_{t-1}^\alpha N_t^{1-\alpha} - W_t N_t - r_t^k K_{t-1}$$

The First-Order Conditions (FOCs) yield factor prices equal to their marginal products:

$$r_t^k = \alpha \frac{Y_t}{K_{t-1}}$$

$$W_t = (1 - \alpha) \frac{Y_t}{N_t}$$

3.2 Household's Problem

Households choose sequences of $\{C_t, N_t, K_t\}$ to maximize expected utility subject to the budget constraint:

$$C_t + K_t - (1 - \delta)K_{t-1} = W_t N_t + r_t^k K_{t-1} + T_t + \Pi_t$$

The Lagrangian yields two fundamental optimality conditions:

1. **Intratemporal Labor-Leisure Choice:** Equating the real wage to the marginal rate of substitution between consumption and leisure:

$$W_t = \frac{\theta C_t}{1 - N_t} \implies N_t = \frac{W_t - \theta C_t}{W_t}$$

2. **Intertemporal Euler Equation:** Dictating optimal consumption smoothing:

$$\frac{1}{C_t} = \beta E_t \left[\frac{1}{C_{t+1}} (r_{t+1}^k + 1 - \delta) \right]$$

4 Steady-State and Calibration

We calibrate the structural parameters $\{\beta, \delta, \theta, \alpha\}$ using steady-state targets. In the steady state ($A_t = \bar{A} = 1$), the Euler equation simplifies to:

$$1 = \beta(\bar{r}^k + 1 - \delta) \implies \bar{r}^k = \frac{1}{\beta} - 1 + \delta$$

Defining $\beta = 1/(1 + \rho)$, we get $\bar{r}^k \approx \rho + \delta$. Using the firm's FOC for capital, the **Capital-to-Output ratio** is:

$$\frac{K}{Y} = \frac{\alpha}{\rho + \delta}$$

The **Investment-to-Output ratio** is derived from the steady-state capital accumulation ($I = \delta K$):

$$\frac{I}{Y} = \delta \left(\frac{K}{Y} \right) = \frac{\alpha \delta}{\rho + \delta}$$

The **Consumption-to-Output ratio** is the residual of the resource constraint:

$$\frac{C}{Y} = 1 - \frac{I}{Y} - \frac{G}{Y} = 1 - \frac{\alpha \delta}{\rho + \delta} - \frac{G}{Y}$$

Finally, assuming households work one-third of their time ($N = 1/3$), we calibrate the **Leisure preference** (θ):

$$\theta = (1 - \alpha) \left(\frac{Y}{C} \right) \left(\frac{1 - N}{N} \right)$$

4.1 Solow Residual Estimation

To calibrate the technology shock process, we isolate Total Factor Productivity (TFP) using empirical data. Taking the natural log of the production function:

$$a_t = y_t - \alpha k_{t-1} - (1 - \alpha)n_t$$

Fitting an AR(1) process to this empirical a_t series yields the persistence ($\rho_a \approx 0.95$) and the standard deviation of the shock (σ_ϵ).

5 Log-Linearization

To solve the model dynamically, we log-linearize the equations around the steady state (where $\hat{x}_t = \ln(X_t) - \ln(\bar{X})$ represents percentage deviations):

$$\text{Resource Constraint: } \hat{y}_t = \frac{C}{Y}\hat{c}_t + \frac{I}{Y}\hat{i}_t + \frac{G}{Y}\hat{g}_t \quad (1)$$

$$\text{Euler Equation: } \hat{c}_t = E_t[\hat{c}_{t+1}] - [1 - \beta(1 - \delta)]E_t[\hat{r}_{t+1}^k] \quad (2)$$

$$\text{Law of Capital: } \hat{i}_t = \frac{1}{\delta}\hat{k}_t - \frac{1 - \delta}{\delta}\hat{k}_{t-1} \quad (3)$$

$$\text{Labor Supply: } \hat{w}_t = \hat{c}_t + \frac{N}{1 - N}\hat{n}_t \quad (4)$$

$$\text{Capital Demand: } \hat{r}_t^k = \hat{y}_t - \hat{k}_{t-1} \quad (5)$$

$$\text{Labor Demand: } \hat{w}_t = \hat{y}_t - \hat{n}_t \quad (6)$$

$$\text{Production Function: } \hat{y}_t = \hat{a}_t + \alpha\hat{k}_{t-1} + (1 - \alpha)\hat{n}_t \quad (7)$$

6 Model Simulation and IRFs

6.1 Positive Technology Shock (Supply Shock)

When a positive shock ($\epsilon_t > 0$) hits A_t :

- **Impact:** The marginal products of labor and capital rise instantly. Firms demand more labor, driving up real wages.
- **Labor Battle:** Households face a positive *wealth effect* (wanting more leisure) and a positive *substitution effect* (leisure is temporarily more expensive due to higher wages). In the RBC model, the substitution effect dominates, causing hours worked (N_t) to rise.
- **Consumption Smoothing:** Households wish to smooth this temporary wealth increase. Consumption (C_t) rises, but much less than output (Y_t). The excess output is aggressively saved, causing Investment (I_t) to spike.
- **Transition:** As the shock decays, the economy is left with an elevated capital stock. Diminishing returns drive the interest rate (r_t^k) below its steady state. Investment eventually turns negative (relative to steady state) to allow the capital stock to depreciate back to equilibrium.

6.2 Positive Government Expenditure Shock (Demand Shock)

When G_t unexpectedly increases:

- **Wealth Effect:** Households anticipate higher future taxes to fund G_t . This negative wealth effect causes them to reduce consumption (C_t) and consume less leisure, thereby increasing hours worked (N_t).
- **Crowding Out:** Because output cannot instantly rise enough to satisfy the government demand, private investment (I_t) is heavily crowded out.
- **Result:** Output (Y_t) rises slightly due to increased labor supply, but Consumption and Investment fall.

7 Model Performance and the Klein Critique

The RBC framework is highly successful at replicating key business cycle facts: it correctly predicts that investment is significantly more volatile than output, and that consumption is smoother than output. It also generates strong procyclicality in C , I , and N .

However, the model fails in labor market dynamics (often known as the **Labor Puzzle**):

- **Volatility:** The model predicts hours worked are relatively stable, whereas empirical data shows high volatility.
- **Correlation (Paul Klein's Critique):** The RBC model predicts a near-perfect correlation (close to 1) between real wages (or labor productivity) and hours worked, because fluctuations are driven entirely by shifts in Labor Demand. In empirical data, this correlation is essentially zero (the Dunlop-Tarshis observation). Adding fiscal or demand shocks that shift Labor Supply helps break this perfect correlation, bringing the model closer to reality.